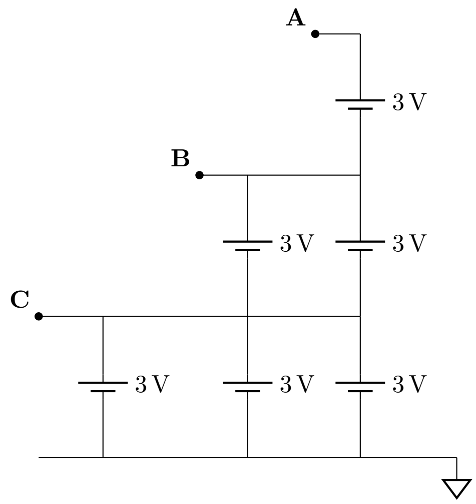
EE 105 — Quiz 1: Example Questions

Module 0 · Fall 2026

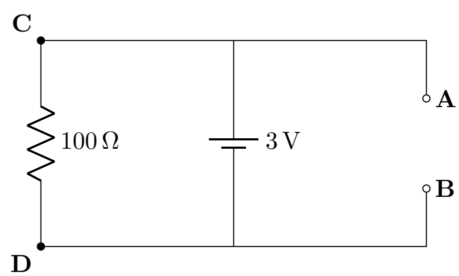
Question 1

Given the circuit with nodes A, B, and C, each connected to 3 V sources as shown, determine the voltages V_A, V_B, and V_C at the respective nodes with respect to ground.



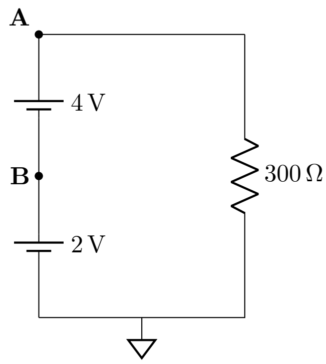
Question 2

Given the circuit with nodes A, B, C, and D as shown, calculate the following: V_{AB}, I_{AB}, and I_{CD}.



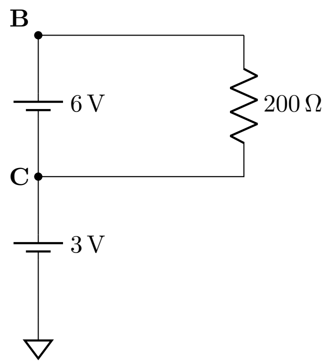
Question 3

Two ideal voltage sources are stacked in series as shown. A 300 Ω resistor is connected from node A to ground. Determine V_B, V_A, and the resistor current I_R (defined from A to ground).



Question 4

The 3 V and 6 V sources are stacked in series as shown. A 200 Ω resistor is connected between nodes B and C. Determine V_C, V_B, V_{BC}, and I_{BC} (defined from B to C).



Solutions

Question 1 Solution

- Node C is one 3 V source above ground, so $V_C = \mathbf{3\,V}$.
- Node B is another 3 V above node C, so $V_B = \mathbf{3\,V + 3\,V = 6\,V}$.
- Node A is another 3 V above node B, so $V_A = \mathbf{6\,V + 3\,V = 9\,V}$.

  Answer: $V_A = 9\,\mathrm{V}$, $V_B = 6\,\mathrm{V}$, $V_C = 3\,\mathrm{V}$.

Question 2 Solution

- A and C are connected by wire, and B and D are connected by wire. The 3 V source therefore makes the top node 3 V above the bottom node: $V_{AB} = \mathbf{3\,V}$.
- The A–B branch is open, so no current can flow in that branch: $I_{AB} = \mathbf{0\,A}$.
- The 100 Ω resistor is directly across 3 V. By Ohm's law, $I_{CD} = \mathbf{3\,V\ /\ 100\,\Omega = 0.03\,A = 30\,mA}$.

  Answer: $V_{AB} = 3\,\mathrm{V}$, $I_{AB} = 0\,\mathrm{A}$, $I_{CD} = 30\,\mathrm{mA}$ (from C to D).

Question 3 Solution

- The first source raises node B by 2 V above ground: $V_B = \mathbf{2\,V}$.
- The 4 V source is stacked on top of the 2 V source, so $V_A = \mathbf{2\,V + 4\,V = 6\,V}$.
- The 300 Ω resistor is connected from A to ground, so it has 6 V across it. By Ohm's law, $I_R = \mathbf{6\,V\ /\ 300\,\Omega = 0.02\,A = 20\,mA}$.

  Answer: $V_B = 2\,\mathrm{V}$, $V_A = 6\,\mathrm{V}$, $I_R = 20\,\mathrm{mA}$ (from A to ground).

Question 4 Solution

- The 3 V source places node C at $V_C = \mathbf{3\,V}$.
- The 6 V source is stacked above it, so $V_B = \mathbf{3\,V + 6\,V = 9\,V}$.
- The voltage from B to C is therefore $V_{BC} = \mathbf{9\,V - 3\,V = 6\,V}$.
- The 200 Ω resistor has 6 V across it. By Ohm's law, $I_{BC} = \mathbf{6\,V\ /\ 200\,\Omega = 0.03\,A = 30\,mA}$.

  Answer: $V_C = 3\,\mathrm{V}$, $V_B = 9\,\mathrm{V}$, $V_{BC} = 6\,\mathrm{V}$, $I_{BC} = 30\,\mathrm{mA}$ (from B to C).